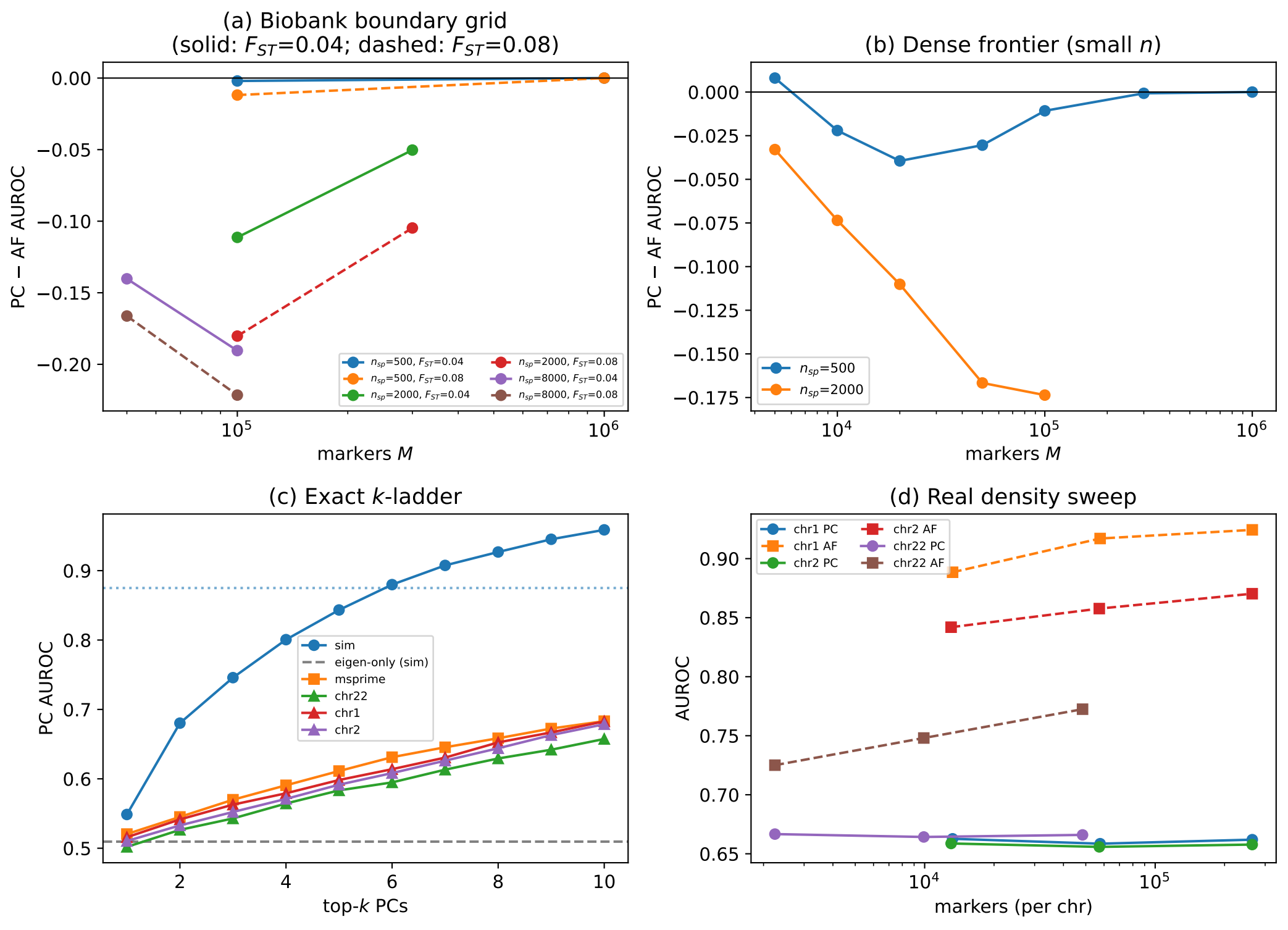
(a) Biobank boundary grid
(solid: F_{ST}=0.04; dashed: F_{ST}=0.08)
PC − AF AUROC
markers M
n_{sp}=500, F_{ST}=0.04
n_{sp}=500, F_{ST}=0.08
n_{sp}=2000, F_{ST}=0.04
n_{sp}=2000, F_{ST}=0.08
n_{sp}=8000, F_{ST}=0.04
n_{sp}=8000, F_{ST}=0.08
(b) Dense frontier (small n)
PC − AF AUROC
markers M
n_{sp}=500
n_{sp}=2000
(c) Exact k-ladder
PC AUROC
top-k PCs
sim
eigen-only (sim)
msprime
chr22
chr1
chr2
(d) Real density sweep
AUROC
markers (per chr)
chr1 PC
chr1 AF
chr2 PC
chr2 AF
chr22 PC
chr22 AF